EE2004: Digital Signal Processing

Tutorial 6

(March 9, 2026)

1. Determine and sketch the DTFT $X_i(\omega)$ of $x_i[n]$:

   (a) $x_1[n] = \{1, 1, \overset{\downarrow}{1}, 1, 1\}$

   (b) $x_2[n] = \{1, 0, 1, 0, \overset{\downarrow}{1}, 0, 1, 0, 1\}$

   (c) $x_3[n] = \{1, 0, 0, 1, 0, 0, \overset{\downarrow}{1}, 0, 0, 1, 0, 0, 1\}$

   Is there any relation between $X_1(\omega)$, $X_2(\omega)$, and $X_3(\omega)$?

2. We are given the following information about an LTI system with frequency response $H(e^{j\omega})$: (a) the system is causal, (b) (b) $H(e^{j\omega}) = H^*(e^{-j\omega})$, and (c) the DTFT of $h[n+1]$ is real-valued. Based on the information given, is the filter IIR or FIR? Justify your answer.

3. Consider a discrete-time LTI system with impulse response $h[n] = (1/2)^n\, u[n]$. Using DTFT properties determine the response to each of the following input signals: (a) $x[n] = (n+1)\,(1/4)^n\, u[n]$, and (b) $x[n] = (-1)^n$.

4. Consider a causal LTI system described by the difference equation

$$y[n] + \frac{1}{2}y[n-1] = x[n]$$

   Determine the frequency response $H(e^{j\omega})$ of the above system.

5. The input to a cascade of two LTI systems is $x[n] = \cos(0.6\pi n) + 3\delta[n-5] + 2$. The first system is an ideal lowpass filter with $\omega_c = \pi/2$. The second system's input-output relationship is $y[n] = w]n] - w[n-1]$, where $w[n]$ is the output of the first system. Find the output $y[n]$.

6. A causal and stable LTI system accepts the input $\left(\frac{4}{5}\right)^n u[n]$ and produces the output $n\left(\frac{4}{5}\right)^n u[n]$.

   (a) Determine the system's frequency response $H(e^{j\omega})$.

   (b) Determine the system's input-output difference equation.

7. Consider a system obtained by cascading two LTI systems with frequency responses

$$H_1(e^{j\omega}) = \frac{2 - e^{-j\omega}}{1 + 0.5e^{-j\omega}}$$

   and

$$H_2(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega} + 0.25e^{-j2\omega}}$$

   (a) Find the difference equation describing the overall system. (b) Determine the impulse response of the overall system.

8. Consider and LTI system with frequency response $H(e^{j\omega}) = \dfrac{1 - e^{-j2\omega}}{1 + r^2 e^{-j2\omega}}$ where $r = 0.9$. The input to this system is $\cos n\pi/8 + \cos n\pi/2$. The output can be expressed as $A_1 \cos(n\pi/8 + \theta_1) + A_2 \cos(n\pi/2 + \theta_2)$. Determine A_i and θ_i for $i = 1, 2$.

9. **Computer assignment** Let $x[n] = \{1, \underset{\downarrow}{2}, 3, 4, 5\}$. Numerically evaluate the sequence's DTFT at 501 equispaced points over the interval $[0, \pi]$.

10. **Computer assignment** Numerically verify the validity of the shift property as follows: (i) create a vector of length 11 with random entries using the `rand` function; (ii) compute its DTFT at 501 equispaced points over the interval $[0, \pi]$; (iii) evaluate numerically $e^{-j2\omega} X(e^{j\omega})$ over the same set of points; (iv) form a shifted sequence by adding two zeros at the beginning of the vector created in (i); (v) compute the shifted vector's DTFT at 501 equispaced points over the interval $[0, \pi]$; (vi) finally compute the maximum absolute error of the difference vector and confirm that the value is extremely small.

11. **Computer assignment** Consider the LTI system specified by the difference equation $y[n] = 0.8\, y[n-1] + x[n]$. Let the input to this system be $x[n] = \cos(0.05\pi n)\, u[n]$.
    (a) Generate 100 samples of the input $x[n]$ and store it in the vector `x`.
    (b) Understand MATLAB's `filter` command and obtain the output `y` using `y = filter(b,a,x);` (the vectors `b` and `a` contain the numerator and denominator coefficients of the rational transfer function $H(e^{j\omega})$).
    (c) Plot `x` and `y` in a figure one below the other (use the command `subplot`). Observe the initial cycles and compare them with the later ones; what are the main differences? Can you explain why the initial part is different?

◇